

# JEE MAIN REVISION MASTER: UNITS & MEASUREMENTS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 227 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                    | Frequency | Category   |
|---------------|-------------------------------------------------|-----------|------------|
| Very High     | Dimensional Analysis & Principle of Homogeneity | 119 PYQs  | Dimensions |
| Very High     | Error Analysis & Significant Figures            | 108 PYQs  | Errors     |

## 2. Master Formula Reference Card

| Concept / Topic                                 | Mathematical Formulation & Governing Equations                                                                                  |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Dimensional Analysis & Principle of Homogeneity | $[G] = M^{-1}L^3T^{-2}$ , $[h] = ML^2T^{-1}$ , $[\mu_0] = MLT^{-2}A^{-2}$                                                       |
| Error Analysis & Significant Figures            | $\frac{\Delta Z}{Z} = a\frac{\Delta A}{A} + b\frac{\Delta B}{B} + c\frac{\Delta C}{C}$<br>$\text{for } Z = \frac{A^a B^b}{C^c}$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Dimensional Analysis & Principle of Homogeneity (Very High Yield — 119 PYQs)

**Core Principle:** Application of dimensions to check equation consistency and derive relations between physical quantities.

■ **Caution / JEE Pitfall:** Trigonometric, logarithmic, and exponential arguments must strictly be dimensionless.

■ **Speed Trick:** Equate powers of M, L, T on both sides to find unknown exponents.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "A new unit (  $\alpha$  ) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Ea..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Dimensions of universal gravitational constant (  $G$  ) in terms of Planck's constant (  $\hbar$  ), distance (  $L$  ), mass (  $M$  ) and time (  $T$  )..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "If  $\epsilon, E$  and  $t$  represent the free space permittivity, electric field and time respectively, then the unit of (  $\epsilon E / t$  )..." [Ans: C]

### 3.2 Error Analysis & Significant Figures (Very High Yield — 108 PYQs)

**Core Principle:** Maximum fractional and percentage errors in combination of quantities and screw gauge / vernier calipers reading.

■ **Caution / JEE Pitfall:** Errors always ADD up in worst-case analysis; never subtract fractional errors.

■ **Speed Trick:** Least count = Pitch / (Number of circular divisions). Total reading = MSR + (CSR × LC) - (Zero Error).

**Representative Source PYQs:**

- **[2026 Morning Shift] Q#1 (MCQ):** "The density  $\rho$  of a uniform cylinder is determined by measuring its mass  $m$ , length  $l$  and diameter  $d$ . The measured values of  $m, l, \dots$ " **[Ans: B]**
- **[2026 Morning Shift] Q#1 (MCQ):** "In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and  $7^{\circ}$ ..." **[Ans: C]**
- **[2026 Evening Shift] Q#1 (MCQ):** "The time period of a simple harmonic oscillator is  $T = 2\pi\sqrt{(k/m)}$ . Measured value of mass ( $m$ ) of the object is 10 g with an a..." **[Ans: D]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A new unit ( $\alpha$ ) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of  $\alpha$  units if light takes 6 min. 40 s to cover this distance?

- (A)  $200\alpha$
- (B)  $400\alpha$  **[Correct]**
- (C)  $300\alpha$
- (D)  $500\alpha$

**Correct Answer: B**

**Solution Analysis:** To find the distance between two points when an object moves at a constant speed, we use formula, Distance = Speed × Time  $\alpha$  is a new unit of length such that 1 unit ( $\alpha$ ) = speed of light in vacuum ( $c$ ). The speed of light  $c$  is the distance light travels per unit of time (usually meters per second).  $1\alpha = c \times (1\text{second})$  The time taken for light to travel from Venus to Earth is 6 minutes and 40 seconds. 1 minute = 60 seconds 6 minutes =  $6 \times 60 = 360$  seconds Total time  $t = 360\text{s} + 40\text{s} = 400$  seconds Using distance formula: Distance = Speed of light × Time Distance  $= c$

**Benchmark Problem 2: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)**

The density  $\rho$  of a uniform cylinder is determined by measuring its mass  $m$ , length  $l$  and diameter  $d$ . The measured values of  $m, l$  and  $d$  are  $97.42 \pm 0.02\text{g}$ ,  $8.35 \pm 0.05\text{mm}$  and  $20.20 \pm 0.02\text{mm}$ , respectively. Calculated percentage fractional error in  $\rho$  is \_\_\_\_.

- (A) 0.63%
- (B) 0.82% **[Correct]**
- (C) 0.72%
- (D) 0.25%

**Correct Answer: B**

**Solution Analysis:** The density ( $\rho$ ) of a body is defined as the ratio of its mass ( $m$ ) to its volume ( $V$ ).  $\rho = (m / V)$  For a uniform cylinder of length  $l$  and diameter  $D$ , the volume is,  $V = \pi r^2 l = \pi ((D / 2))^2 l = \frac{\pi D^2 l}{4}$  Substituting the volume into the density formula:  $\rho = (m / V) = \frac{4m}{\pi D^2 l}$  To find the fractional error, we take the natural logarithm ( $\ln$ ) on both sides:  $\ln(\rho) = \ln(4) + \ln(m) - \ln(\pi) - 2\ln(D) - \ln(l)$  Differentiating both sides to find the relative change  $(d\rho / \rho) = (dm / m) - (2dD / D) - (dl / l)$  The maximum possible fractional e